



Sustainability and Environmental Assessment (A25)

LIFE CYCLE ASSESSMENT OF BEER PACKAGING IN THE UK

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Life Cycle Assessment of Beer Packaging in the UK

A Comparative Screening Life Cycle Assessment of Glass Bottles and Aluminium Cans

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1. Introduction

The environmental sustainability of food and beverage systems has gained increasing attention due to their substantial contributions to global greenhouse gas (GHG) emissions, energy consumption, and material use. In the United Kingdom, beer is one of the most widely consumed alcoholic beverages, with production occurring at large industrial scales and distribution networks extending nationwide. While brewing processes themselves require significant energy and water inputs, packaging and distribution stages are increasingly recognised as critical contributors to the overall life cycle environmental footprint of beer products.

Glass bottles remain a common packaging format for beer in the UK, particularly within the off-trade sector. Glass offers advantages such as chemical inertness and recyclability; however, it is highly material-intensive and energy-intensive to manufacture. In contrast, aluminium cans are lightweight, highly recyclable, and efficient to transport, making them an increasingly popular alternative. Comparing the environmental performance of these two packaging formats is therefore of practical relevance for sustainability decision-making within the brewing sector.

Life Cycle Assessment (LCA) is a standardised environmental assessment methodology used to quantify environmental impacts across the life cycle of a product or service, from raw material extraction through production and distribution (ISO, 2006a; ISO, 2006b). By applying LCA, it is possible to identify emission hotspots and evaluate trade-offs between alternative packaging systems under consistent assumptions.

The aim of this study is to conduct a screening life cycle carbon footprint assessment of beer production and distribution in the UK, focusing on a 330 mL serving size. Two packaging scenarios are assessed: a single-use glass bottle and an aluminium can. The objectives are to (i) quantify greenhouse gas emissions expressed as Global Warming Potential (GWP), (ii) identify dominant contributors across the life cycle, and (iii) compare the environmental performance of glass and aluminium packaging formats.

2. Methodology

2.1 Assessment approach

This study applies a screening Life Cycle Assessment, following the principles outlined in ISO 14040 and ISO 14044. A screening approach is appropriate where the objective is to identify major contributors and compare alternative scenarios using a transparent but limited dataset. The assessment focuses on a single impact category - Global Warming Potential (GWP) - expressed in kilograms of carbon dioxide equivalent (kg CO₂e).

All modelling and calculations were implemented using a spreadsheet-based tool (Microsoft Excel), enabling explicit tracking of assumptions, activity data, emission factors, and unit conversions.

2.2 Goal and scope definition

The goal of the study is to estimate and compare the carbon footprint of beer packaged in a glass bottle and an aluminium can in the UK and to identify opportunities for reducing life cycle emissions.

The functional unit (FU) is defined as:

One 330 mL unit of beer delivered to the consumer in the UK.

This functional unit ensures equivalence between packaging formats and reflects a typical consumption unit.

The system boundary as shown in **Figure 1** is defined as cradle-to-distribution, including brewing energy use, packaging production, and refrigerated road transport to retail. Excluded stages include agricultural production of raw ingredients, retail refrigeration, consumer use, and end-of-life waste management. An overview of the goal, functional unit, and system boundary is provided in **Table 1**.

3. Life Cycle Inventory (LCI)

The life cycle inventory combines primary data collected through direct measurement with secondary data sourced from literature and official datasets. The inventory data per functional unit are summarised in **Table 2**.

3.1 Brewing energy

Brewing energy consumption was represented using published UK brewery energy intensities. Electricity use was assumed to be 8 kWh per hectolitre of beer, while thermal energy demand (natural gas) was assumed to be 25 kWh per hectolitre. These values are representative of modern industrial brewing operations and reflect the heat-intensive nature of brewing processes such as wort boiling and cleaning (Amienyo & Azapagic, 2016).

3.2 Packaging materials

Packaging masses were primarily obtained through direct measurement. Empty glass bottles and aluminium cans were weighed using a kitchen scale and averaged. The measured glass bottle mass was 0.25 kg per bottle, while the aluminium can mass was 0.015 kg per empty can. Cardboard packaging was also measured and allocated on a per-unit basis. For aluminium cans, printed graphics eliminated the need for paper labels, and no crown caps were required.

3.3 Transport

A brewery-to-retail transport distance of 300 km was assumed, consistent with UK beer distribution modelling approaches reported by Amienyo and Azapagic (2016). Transport activity was modelled using tonne-kilometres (tonne-km), calculated from the combined mass of beer and packaging. Refrigerated heavy goods vehicle (HGV) transport was assumed for all distribution stages to represent a conservative cold-chain scenario.

4. Life Cycle Impact Assessment

Emission factors used in the impact assessment are presented in **Table 3**.

Electricity, natural gas, and transport emission factors were sourced from the UK Government Greenhouse Gas Conversion Factors (2025) (Department for Energy Security and Net Zero, 2025). Packaging material emission factors were sourced from CarbonCloud, supplemented by literature-based Environmental Product Declarations (EPDs). For aluminium packaging, a cradle-to-gate emission factor of 13.9 kg CO₂e per kg was applied, representing a conservative assumption with relatively low recycled content.

Global Warming Potential was calculated using:

$$\text{GWP} = \text{Activity} \times \text{Emission Factor}$$

5. Results

5.1 Glass bottle scenario

The total carbon footprint of a 330 mL glass-bottled beer was estimated at 0.326 kg CO₂e per unit. As shown in **Table 4**, packaging dominated life cycle emissions, with the glass bottle alone contributing approximately 77% of total emissions. Brewing energy contributed a smaller share, while refrigerated transport accounted for approximately 7% of the total.

5.2 Aluminium can scenario

As shown in **Table 5**, the aluminium can scenario resulted in a total carbon footprint of 0.248 kg CO₂e per 330 mL unit. Aluminium production dominated emissions in this scenario, accounting for approximately 84% of total emissions. Despite the high emission factor per kilogram of aluminium, the substantially lower packaging mass resulted in a lower overall footprint than the glass bottle. Transport-related emissions were also lower due to the reduced total mass.

5.3 Comparative analysis

A comparison of the two packaging formats indicates that the aluminium can has a 24% lower carbon footprint than the single-use glass bottle under identical brewing energy and transport assumptions. This comparison is summarised in **Table 6**.

6. Discussion

6.1 Interpretation of findings

The results show that packaging is the dominant contributor to the life cycle carbon footprint of beer in both scenarios (Amienyo & Azapagic, 2016; Kafkova & Weinzettel, 2025). In the glass bottle case, the high mass of container glass relative to the beer volume leads to a disproportionately large contribution to total emissions (Kafkova & Weinzettel, 2025). By contrast, the aluminium can scenario demonstrates that a lightweight packaging format can achieve a lower overall carbon footprint despite higher emissions per kilogram of material (Marrucci et al., 2024; Kafkova & Weinzettel, 2025). This highlights that material intensity is a more influential driver than emission factors alone when comparing packaging systems (Marrucci et al., 2024).

Brewing energy contributed a secondary share of emissions in both scenarios (Amienyo & Azapagic, 2016). While energy efficiency improvements remain important, the results indicate that packaging choices offer greater potential for reducing product-level emissions than incremental changes in brewing energy (Amienyo & Azapagic, 2016; Marrucci et al., 2024). Transport emissions were present but not dominant, even under conservative refrigerated transport assumptions.

6.2 Sensitivity and uncertainty

Sensitivity analysis (**Table 7**) indicates that results are most sensitive to packaging mass and material emission factors, particularly for aluminium (Marrucci et al., 2024; Kafkova &

Weinzettel, 2025). Aluminium emission factors vary widely depending on recycled content and electricity sources used in smelting (Marrucci et al., 2024). Increasing recycled aluminium content would significantly reduce the carbon footprint of the can scenario and further enhance its advantage relative to glass (Kafkova & Weinzettel, 2025). Although transport assumptions introduce some uncertainty, variations are unlikely to alter the relative ranking of packaging formats (Kafkova & Weinzettel, 2025).

6.3 Limitations

This study relies partly on secondary emission factor data and excludes end-of-life recycling benefits, which could reduce net emissions for both packaging formats (Amienyo & Azapagic, 2016; Kafkova & Weinzettel, 2025). Agricultural production impacts were also excluded, potentially underestimating total life cycle emissions (Amienyo & Azapagic, 2016). Nevertheless, the screening LCA approach is appropriate for identifying emission hotspots and comparing packaging alternatives under consistent assumptions (ISO, 2006a; ISO, 2006b).

7. Conclusions and Recommendations

7.1 Conclusions

This screening LCA estimated the carbon footprint of a 330 mL glass-bottled beer at 0.326 kg CO₂e and that of an aluminium-canned beer at 0.248 kg CO₂e. Packaging was identified as the dominant contributor in both scenarios. The aluminium can exhibits a 24% lower carbon footprint than the glass bottle, primarily due to its lower packaging mass and reduced transport burden.

7.2 Recommendations

1. Adopt lightweight packaging formats, particularly aluminium cans where product requirements allow.
2. Increase recycled content in both glass and aluminium packaging.
3. Explore returnable glass systems where feasible.
4. Improve thermal energy efficiency in brewing operations.
5. Optimise logistics, including reducing unnecessary refrigerated transport.

8. Reflection on Personal Development

This project improved my understanding of life cycle assessment through the development of a transparent Excel-based screening LCA model. Constructing the life cycle inventory and comparing glass bottles with aluminium cans strengthened my ability to interpret environmental data and link numerical results to sustainability decisions.

The main challenge was managing uncertainty due to limited primary data, which required several assumptions, particularly for transport and average distances. Addressing this through sensitivity analysis highlighted the importance of transparency in LCA studies.

If repeating the project, I would prioritise earlier data collection and consider extending the system boundary to include end-of-life processes. Overall, the work enhanced my technical skills and critical understanding of environmental assessment methods.

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Appendix

Appendix A

List of Tables

Table 1

Goal, functional unit, and system boundary of the life cycle assessment

Element	Description
Goal of the study	To quantify and compare the greenhouse gas emissions of beer packaged in glass bottles and aluminium cans in the UK
Functional unit	One 330 mL unit of beer delivered to the consumer
Impact category	Global Warming Potential (GWP, kg CO ₂ e)
System boundary	Cradle-to-distribution
Included processes	Brewing energy (electricity and gas), packaging production, secondary packaging, refrigerated road transport
Excluded processes	Agriculture, retail refrigeration, consumer use, end-of-life

Table 2

Life cycle inventory data per 330 mL unit of beer

Parameter	Glass bottle	Aluminium can	Unit	Data source
Beer volume	0.33	0.33	L/unit	Assumed
Beer mass	0.33	0.33	kg/unit	Density assumption
Electricity use	0.0264	0.0264	kWh/unit	Literature (UK breweries)
Thermal energy (gas)	0.0825	0.0825	kWh/unit	Literature (UK breweries)
Primary packaging mass	0.25 (glass)	0.015 (aluminium)	kg/unit	Measured
Crown cap	0.0025	0	kg/unit	Measured / not required
Label	0.001	0	kg/unit	Assumed
Cardboard	0.02	0.01	kg/unit	Measured
Plastic film	0	0	kg/unit	Negligible
Transport distance	300	300	km	Assumed
Transport activity	0.181	0.1065	tonne-km/unit	Calculated

Table 3*Emission factors applied in the life cycle impact assessment*

Flow	Emission factor	Unit	Source
Electricity (UK grid)	0.177	kg CO ₂ e/kWh	UK GHG Conversion Factors (2025)
Natural gas	0.18296	kg CO ₂ e/kWh	UK GHG Conversion Factors (2025)
Glass packaging	1.0	kg CO ₂ e/kg	Literature / EPD
Aluminium packaging	13.9	kg CO ₂ e/kg	CarbonCloud
Steel/aluminium cap	7.67	kg CO ₂ e/kg	CarbonCloud
Paper label	1.56	kg CO ₂ e/kg	CarbonCloud
Cardboard	0.71	kg CO ₂ e/kg	CarbonCloud
Refrigerated HGV transport	0.11894	kg CO ₂ e/tonne-km	UK GHG CF (2025)

Table 4*Greenhouse gas emissions for the glass bottle scenario*

Life cycle stage	Impact (kg CO ₂ e/unit)	Contribution (%)
Electricity	0.0047	1.4
Natural gas	0.0151	4.6
Glass bottle	0.25	76.7
Cap	0.0192	5.9
Label	0.0016	0.5
Cardboard	0.0142	4.4
Transport	0.0215	6.6
Total	0.326	100

Table 5*Greenhouse gas emissions for the aluminium can scenario*

Life cycle stage	Impact (kg CO ₂ e/unit)	Contribution (%)
Electricity	0.0047	1.9
Natural gas	0.0151	6.1
Aluminium can	0.2085	84.1
Cardboard	0.0071	2.9
Transport	0.0127	5.1
Total	0.248	100

Table 6*Comparison of carbon footprints for glass bottles and aluminium cans*

Packaging format	Total impact (kg CO₂e/unit)	Difference vs glass	Reduction
Glass bottle	0.326	–	–
Aluminium can	0.248	–0.078	24%

Table 7*Comparison of carbon footprints for glass bottles and aluminium cans*

Parameter varied	Low case	Base case	High case
Aluminium EF (kg CO ₂ e/kg)	8	13.9	16
Can footprint (kg CO ₂ e)	~0.18	0.248	~0.30
Relative ranking	Can << Glass	Can < Glass	Similar

Appendix B

List of Figures

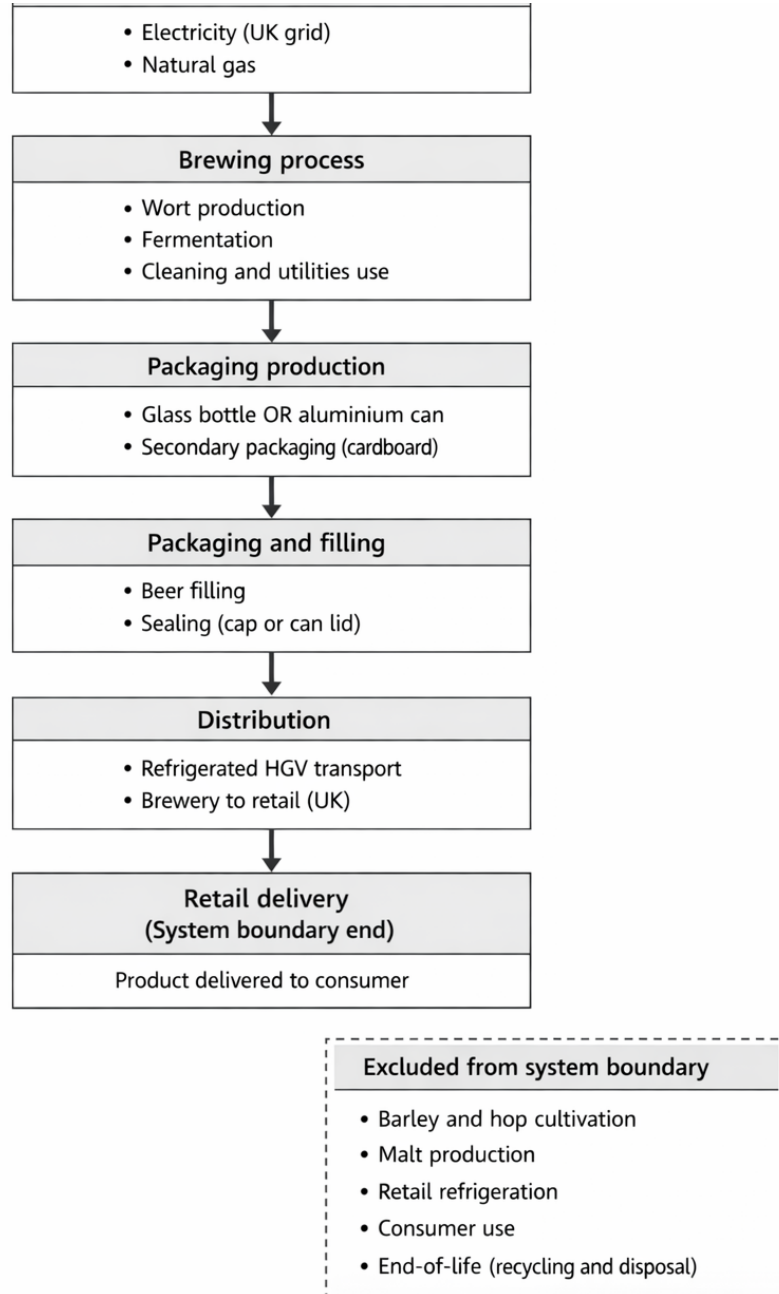


Figure 1

System Boundary and Life Cycle Stages Considered in the Screening Life Cycle Assessment of Beer Packaging

Note. The system boundary represents a cradle-to-distribution assessment and excludes agricultural production, retail refrigeration, consumer use, and end-of-life stages.

Excel Workbook Snapshots

	A	B	C	D
1	Flow	Emission Factor	Unit	Source
2	Electricity (UK grid)	0.177	kgCO2e/kWh	UK GHG Conversion Factors 2025
3	Natural gas	0.18296	kgCO2e/kWh	UK GHG Conversion Factors 2025
4	Glass	1	kgCO2e/kg	Literature / EPD (container glass)
5	Steel/Aluminium (cap)	7.67	kgCO2e/kg	Carbon Cloud
6	Paper (label)	1.56	kgCO2e/kg	Carbon Cloud
7	Cardboard	0.71	kgCO2e/kg	Carbon Cloud
8	Plastic film	0	kgCO2e/kg	
9	Road freight (HGV)	0.11894	kgCO2e/tonne-km	UK GHG CF 2025 - refrigerated HGV
10	Virgin Aluminium can	13.9	kgCO2e/kg	Carbon Cloud
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Model Overview EmissionFactors Assumptions_Bottle Inventory_Bottle Results_Bottle Assumptions_Can Inventory_Can Results_Can Comparison +

Figure 2

Emission Factors

	A	B	C	D	E
1	Parameter	Value	Unit	Notes	
2	FU volume	0.33	L/bottle	330 mL glass bottle	
3	Beer mass	0.33	kg/bottle	Approx. 1 kg/L	
4	Electricity use	8	kWh/hL	Brewing electricity	
5	Thermal energy (gas)	25	kWh/hL	Brewing heat	
6	Glass mass per bottle	0.25	kg/bottle	measured different brand bottles on on kitchen scale and took average	
7	Crown cap mass	0.0025	kg/bottle	measured on kitchen scale	
8	Label mass	0.001	kg/bottle	assumed	
9	Cardboard per bottle	0.02	kg/bottle	measured on kitchen scale and divided total value by number of bottles in one carton	
10	Plastic wrap per bottle	0	kg/bottle	negligible	
11	Total transport distance	300	km	Brewery to distributor= 100-200km, distributor to retail= 50-150km, total range = 150-300km	
12	Electricity per bottle	0.0264	kWh/bottle	Auto-calculated	
13	Gas per bottle	0.0825	kWh/bottle	Auto-calculated	
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Model Overview EmissionFactors Assumptions_Bottle Inventory_Bottle Results_Bottle Assumptions_Can Inventory_Can Results_Can Comparison +

Figure 3

Assumptions for Beer Bottle

	A	B	C
1	Flow	Activity	Unit
2	Electricity	0.0264	kWh/bottle
3	Natural gas	0.0825	kWh/bottle
4	Glass	0.25	kg/bottle
5	Cap	0.0025	kg/bottle
6	Label	0.001	kg/bottle
7	Cardboard	0.02	kg/bottle
8	Plastic film	0	kg/bottle
9	Transport (tonne-km)	0.18105	tonne-km/bottle
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Model_Overview | EmissionFactors | Assumptions_Bottle | **Inventory_Bottle** | Results_Bottle | Assumptions_Can | Inventory_Can | Results_Can | Comparison | +

Figure 4

Inventory for Beer Bottle

	A	B	C	D	E
1	Flow	Activity	EF	Impact (kg CO2e/bottle)	GHG Contribution %
2	Electricity	0.0264	0.177	0.0046728	1.43
3	Natural gas	0.0825	0.18296	0.0150942	4.63
4	Glass	0.25	1	0.25	76.63
5	Cap	0.0025	7.67	0.019175	5.88
6	Label	0.001	1.56	0.00156	0.48
7	Cardboard	0.02	0.71	0.0142	4.35
8	Plastic film	0	0	0	0.00
9	Transport	0.18105	0.11894	0.021534087	6.60
10	TOTAL			0.326236087	100.00
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Model_Overview | EmissionFactors | Assumptions_Bottle | Inventory_Bottle | **Results_Bottle** | Assumptions_Can | Inventory_Can | Results_Can | Comparison | +

Figure 5

Results for Beer Bottle

	A	B	C	D
1	Parameter	Value	Unit	Notes
2	FU volume	0.33	L/bottle	330 mL glass bottle
3	Beer mass	0.33	kg/bottle	Approx. 1 kg/L
4	Electricity use	8	kWh/hL	Brewing electricity
5	Thermal energy (gas)	25	kWh/hL	Brewing heat
6	Aluminium mass per can	0.015	kg/bottle	measured on kitchen scale
7	Crown cap mass	0	kg/bottle	measured on kitchen scale
8	Label mass	0	kg/bottle	assumed
9	Cardboard per can	0.01	kg/bottle	measured on kitchen scale and divided total value by number of cans in one carton
10	Plastic wrap per can	0	kg/bottle	negligible
11	Total transport distance	300	km	Brewery to distributor= 100-200km, distributor to retail= 50-150km, total range = 150-300km
12	Electricity per can	0.0264	kWh/bottle	Auto-calculated
13	Gas per can	0.0825	kWh/bottle	Auto-calculated
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Figure 6

Assumptions for Beer Can

	A	B	C	D
1	Flow	Activity	Unit	
2	Electricity	0.0264	kWh/can	
3	Natural gas	0.0825	kWh/can	
4	Aluminium	0.015	kg/can	
5	Cap	0	kg/can	
6	Label	0	kg/can	
7	Cardboard	0.01	kg/can	
8	Plastic film	0	kg/can	
9	Transport (tonne-km)	0.1065	tonne-km/can	
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Figure 7

Inventory for Beer Can

	A	B	C	D	E
1	Flow	Activity	EF	Impact (kg CO ₂ e/can)	GHG Contribution %
2	Electricity	0.0264	0.177	0.0046728	1.88
3	Natural gas	0.0825	0.18296	0.0150942	6.09
4	Aluminium	0.015	13.9	0.2085	84.06
5	Cap	0	7.67	0	0.00
6	Label	0	1.56	0	0.00
7	Cardboard	0.01	0.71	0.0071	2.86
8	Plastic film	0	0	0	0.00
9	Transport	0.1065	0.11894	0.01266711	5.11
10	TOTAL			0.24803411	100.00
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23	Model_Overview	EmissionFactors	Assumptions_Bottle	Inventory_Bottle	Results_Bottle
			Assumptions_Can	Inventory_Can	Results_Can
				Comparison	+

Figure 8

Results for Beer Can

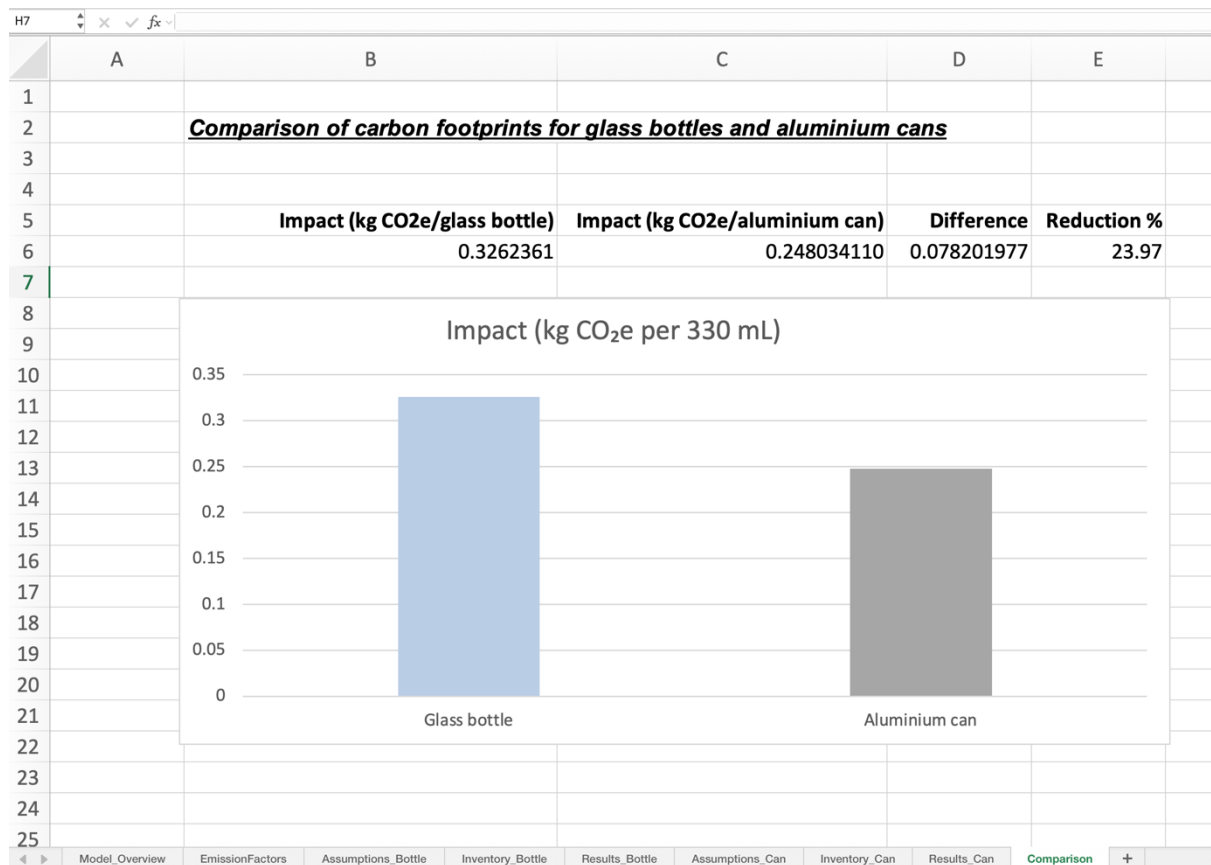


Figure 9

Comparison of Carbon Footprints between Beer Bottle and Beer Can